The Mirage of Significance: How Non-Independence Inflates Predictor Effects in Combinatorial LLM Evaluation Panels

Anonymous

Abstract

Multi-judge LLM evaluation panels are increasingly adopted to improve reliability, yet their vulnerability to adversarial manipulation remains poorly understood. We conduct a large-scale study decomposing panel vulnerability into individual judge weakness ($\eta_{\max}$) and structural diversity (K_{eff}) across 15 models, 455 panels, 9 attack types, and 3 benchmarks. Our central finding is *methodological*: standard OLS regression dramatically overstates predictor significance due to structural non-independence among combinatorial panels—each of 15 models appears in 91 of 455 panels. Under wild cluster bootstrap correction, $\eta_{\max}$ significance drops from 11/12 to 3/12 conditions, and K_{eff} from 8/12 to 1/12, with SE inflation of 3.2×. This collapse scales with panel size: at K=9, SE inflation reaches 7.2× and $\eta_{\max}$ significance falls to 1/12. We further show that multicollinearity (VIF > 7) prevents reliable separation of $\eta_{\max}$ and K_{eff}, and that a BFT-inspired random-fault contrast fails its GO/NO-GO gate. Two findings survive correction: (1) aggregation rule choice fundamentally modulates vulnerability predictability (threshold R^2=0.89 vs. Bayesian R^2=0.10), and (2) selecting minimum-$\eta_{\max}$ panels wins 9/12 conditions regardless of correction method. We release a 15-model, 455-panel adversarial evaluation dataset and provide a non-independence correction framework for combinatorial evaluation designs.

1 Introduction

Multi-judge evaluation panels—where multiple LLMs independently assess candidate outputs and their judgments are aggregated—have emerged as a practical approach to improving evaluation reliability. An implicit assumption in this paradigm is that *diversity* among judges provides robustness, analogous to Byzantine Fault Tolerance (BFT) in distributed systems: uncorrelated judges should resist correlated failures.

We set out to test this assumption at scale by decomposing panel vulnerability under adversarial attack into two components: (1) the individual vulnerability of the weakest judge ($\eta_{\max}$), and (2) the effective number of independent judgment dimensions (K_{eff}, derived from eigenvalue decomposition of inter-judge mutual information matrices). Our study covers 15 models spanning 6 families and 3 capability tiers, $\binom{15}{3} = 455$ three-judge panels, 9 attack types, and 3 benchmarks (MMLU, ARC, MT-Bench).

The headline result is *not* what we expected to find. Under standard OLS regression, $\eta_{\max}$ appears significant in 11/12 conditions and K_{eff} in 8/12—suggesting that individual weakness dominates but diversity also matters. However, the 455 panels are structurally non-independent: each model participates in $\binom{14}{2} = 91$ panels. When we apply wild cluster bootstrap (B=9999, Rademacher weights, 3-position max-p), $\eta_{\max}$ significance collapses to 3/12 and K_{eff} to 1/12. Standard errors inflate by 3.2×, and this inflation *worsens* with panel size (reaching 7.2× at K=9).

This OLS-to-bootstrap collapse is itself a finding of general interest. Combinatorial panel designs—evaluating all $\binom{N}{K}$ subsets of N available models—are a natural experimental paradigm for studying panel composition effects, but the resulting non-independence is structural and severe.

We demonstrate that neither $\eta_{\max}$ nor K_{eff} can be reliably established as independent predictors under corrected inference, and that a VIF analysis (VIF > 7 for both predictors) confirms they share 67% of their explained variance.

Two findings are robust to correction. First, the aggregation rule gradient—threshold voting yields R^2=0.89 while Bayesian calibration yields R^2=0.10—is a descriptive statistic unaffected by SE correction. Second, the practical recommendation to select minimum-$\eta_{\max}$ panels holds in 9/12 conditions under direct comparison, as panel ranking does not depend on regression significance.

Our contributions are: (1) a methodological framework for diagnosing and correcting non-independence in combinatorial evaluation designs; (2) empirical demonstration that OLS significance is inflated by 3.2–7.2× in such designs; (3) a 15-model, 455-panel adversarial evaluation dataset across 9 attack types; and (4) practical guidance on aggregation rule selection and panel composition.

2 Related Work

LLM-as-Judge. Using LLMs as evaluators has become standard practice, with single-judge and multi-judge configurations studied extensively. Panel-based evaluation has been proposed to mitigate individual judge biases, but systematic analysis of panel vulnerability under adversarial conditions remains limited.

Adversarial Attacks on LLM Judges. Prior work has demonstrated that LLM judges are susceptible to prompt injection, sycophancy manipulation, and position bias exploitation. These studies focus on individual judges rather than panel-level vulnerability.

Byzantine Fault Tolerance. BFT provides a theoretical framework for reasoning about panel robustness: a system of n nodes tolerates $f < n/3$ arbitrary (Byzantine) faults. This analogy has been informally invoked in the LLM evaluation literature but never empirically tested.

Non-Independence in Panel Data. Clustered standard errors and bootstrap methods for panel data with cross-sectional dependence are well-established in econometrics. However, their application to combinatorial experimental designs in NLP—where observations are constructed from overlapping subsets of a fixed pool—has received little attention.

3 Method

3.1 Experimental Design

Model Pool. We use 15 LLMs spanning 6 families (OpenAI, Anthropic, Google, Meta, Qwen, Mistral) and 3 capability tiers: frontier (3 models), strong (3 models), and mid-weak (9 models).

Panel Construction. For panel size K=3, we enumerate all $\binom{15}{3} = 455$ panels. For sensitivity analysis, we also examine $K \in \{5, 7, 9\}$, yielding 3,003 to 6,435 panels.

Attack Types. We implement 9 adversarial attacks in two categories:

- **Targeted** (exploit judge-specific weaknesses): prompt injection, score manipulation, position bias, few-shot hijacking, persona manipulation

- **Shared-bias** (exploit common cognitive biases): sycophancy, verbosity bias, authority bias, instruction override

Benchmarks. MMLU (4-choice MCQ), ARC (multiple-choice), and MT-Bench (open-ended pairwise comparison).

Vulnerability Metric. For each panel × attack condition, we measure the *attack success rate* (ASR): the fraction of evaluation instances where the adversarial manipulation changes the panel's aggregated judgment.

3.2 Predictor Variables

Individual Weakness ($\eta_{\max}$). The maximum individual ASR among the K judges in a panel, representing the vulnerability of the weakest link.

Effective Diversity (K_{eff}). Derived from eigenvalue decomposition of the inter-judge mutual information (MI) matrix. K_{eff} captures the effective number of independent judgment dimensions, ranging from 1 (all judges identical) to K (fully independent).

Regression Specification. We use a log-log OLS specification:

$$\log(\text{ASR} + \epsilon) = \beta_0 + \beta_1 \log(\eta_{\max} + \epsilon) + \beta_2 \log(K_{\text{eff}}) + \varepsilon \tag{1}$$

3.3 Non-Independence Correction Methods

The $\binom{N}{K}$ combinatorial design creates severe non-independence: each model appears in $\binom{N-1}{K-1}$ panels. We apply four correction methods:

1. **OLS** (naive baseline): assumes independent observations.
2. **Clustered SE** (CR1): clusters on judge identity (15 clusters).
3. **Mixed effects**: random intercepts for each judge.
4. **Wild cluster bootstrap v2**: B=9999 replications, Rademacher weights, max-p across 3 cluster positions. Most conservative.

4 Experiments and Results

4.1 Scale Expansion and Significance Collapse

Table 1 presents our central result. Under OLS, $\eta_{\max}$ appears significant in 11/12 conditions and K_{eff} in 8/12 (7 positive, 1 negative), suggesting a clear dominance pattern. However, all three correction methods substantially reduce significance counts.

The bootstrap is the most conservative method: $\eta_{\max}$ retains significance in only 3/12 conditions, all on MMLU (2/6 MMLU, 1/6 ARC). K_{eff} survives in only 1/12. The dominance ratio ($\eta_{\max}$:K_{eff} significant conditions) actually *increases* from 1.4:1 (OLS) to 3:1 (bootstrap)—but both predictors are near floor.

Table 2 shows per-condition detail. The most striking case is ARC × prompt injection: OLS $p = 3 \times 10^{-14}$ collapses to bootstrap $p = 0.35$, indicating that the OLS signal was almost entirely an artifact of panel overlap.

Table 1: OLS vs. correction methods: significance counts across 12 conditions (MMLU × 6 attacks + ARC × 6 attacks, $p < 0.05$).

Method	$\eta_{\max}$ sig	K_{eff} sig	K_{eff} +/−	Dom. ratio
OLS (naive)	11/12	8/12	7/1	1.4:1
Clustered SE (CR1)	5/12	0/12	—	∞
Mixed effects	4/12	1/12	1/0	4:1
Wild cluster bootstrap v2	3/12	1/12	1/0	3:1
SE inflation factor		3.2×		

Table 2: Per-condition bootstrap results (selected conditions).

Dataset	Attack	OLS η p	Boot η p	OLS K p	Boot K p
MMLU	prompt_inj	2e-6	0.028	0.004	0.18
MMLU	score_manip	8e-5	0.041	0.011	0.31
MMLU	position_bias	0.001	0.08	0.035	0.04
ARC	prompt_inj	3e-14	0.35	0.001	0.62
ARC	score_manip	1e-8	0.06	0.003	0.44
ARC	sycophancy	0.013	0.52	0.24	0.71

Bootstrap significance varies by cluster position: position 0 (most imbalanced, 91:1 panel ratio) yields 2/12, while positions 1 and 2 yield 5/12 and 4/12 respectively. The 3-position max-p (3/12) is conservative but legitimate.

4.2 Attack Expansion

We extend to 9 attack types (adding few-shot hijacking, persona manipulation, instruction override), creating 18 total conditions. The pattern is consistent: OLS significance (16/18 $\eta_{\max}$, 11/18 K_{eff}) collapses under bootstrap (4/18 and 1/18 respectively), confirming that the non-independence problem is structural.

Table 3: New attack types: OLS vs. bootstrap correction (6 conditions).

Dataset	Attack	OLS η p	Boot η p	OLS K p	Boot K p
MMLU	few-shot-hijack	0.001	0.18	0.02	0.41
MMLU	persona-manip	0.003	0.09	0.04	0.52
MMLU	instr-override	5e-4	0.04	0.08	0.38
ARC	few-shot-hijack	0.008	0.31	0.05	0.55
ARC	persona-manip	0.012	0.22	0.09	0.63
ARC	instr-override	0.002	0.12	0.11	0.48

The attack expansion also degrades the GAM-based attack taxonomy. The original 6-attack, 6:6 targeted/shared-bias split (all targeted showing positive K_{eff}, all shared-bias showing negative) develops 3 reversals at 9 attacks (Table 4). Position bias reverses direction on ARC, and verbosity bias reverses on MMLU. The taxonomy retains a directional tendency (∼60–70% agreement) but is no longer clean.

Table 4: Extended GAM taxonomy: K_{eff} direction across 9 attacks × 2 benchmarks.

Attack	Category	K_{eff} dir (MMLU)	K_{eff} dir (ARC)	Consistent?
prompt_injection	targeted	+	+	yes
score_manipulation	targeted	+	+	yes
position_bias	targeted	+	–	**REVERSAL**
few-shot-hijacking	targeted	+	+	weak
persona-manipulation	targeted	+	+	weak
sycophancy	shared-bias	–	–	yes
verbosity_bias	shared-bias	–	+	**REVERSAL**
authority_bias	shared-bias	–	–	yes
instruction-override	shared-bias	–	–	yes

4.3 BFT Random-Fault Contrast

We test whether K_{eff} differentiates random faults from adaptive attacks—the core BFT-inspired hypothesis. Under random faults (each judge fails independently with probability η_j, majority vote), K_{eff} achieves higher R^2 (0.52 vs. 0.25 for adaptive) but collapses under bootstrap correction (0/2 significant). The GO/NO-GO gate fails on all criteria (Table 5).

Table 5: BFT contrast: effect size analysis. All metrics fail their thresholds.

Metric	Value	Threshold	Verdict
K_{eff} β ratio (RF/adaptive)	1.2	> 2.0	FAIL
Cohen's d (paired)	0.12	> 0.2	FAIL
Contrast p-value	0.31	< 0.05	FAIL
R^2 gap (RF – adaptive)	0.27	> 0.3	MARGINAL

The failure is not because BFT theory is wrong, but because corrected inference cannot reliably isolate K_{eff}'s independent contribution in either regime. With only 15 model clusters and VIF > 7, the regression framework is fundamentally underpowered for predictor-level decomposition.

4.4 Capability-Matched Validation

Within-tier analysis (3 frontier, 3 strong, 9 mid-weak models) reveals that $\eta_{\max}$ dominance is contingent on cross-tier panel composition. When panels are restricted to capability-matched models, $\eta_{\max}$ range compresses by 75% and significance drops dramatically (Table 6).

Table 6: Within-tier vs. cross-tier regression significance.

Scope	N panels	$\eta_{\max}$ sig	K_{eff} sig	$\eta_{\max}$ IQR
Frontier (3 models)	1	0/12	0/12	0.02
Strong (3 models)	1	0/12	0/12	0.04
Mid-weak (9 models)	84	3/12	0/12	0.11
Cross-tier (15 models)	455	11/12	8/12	0.38

This demonstrates that $\eta_{\max}$ significance in the full pool reflects capability tier spread rather than a general weakest-link principle. When panels contain models of similar capability, neither predictor is informative.

4.5 VIF and Multicollinearity Analysis

Variance inflation factor analysis confirms that $\eta_{\max}$ and K_{eff} cannot be reliably separated as independent predictors (Table 7).

Table 7: Multicollinearity diagnostics and variance decomposition.

Diagnostic	$\eta_{\max}$	K_{eff}	Threshold	Verdict
VIF (with tier indicators)	7.8	6.9	> 5	FAIL
Partial r (given other)	0.31	0.12	—	K_{eff} weak
Unique variance %	18%	3%	—	K_{eff} negligible
Condition number	18.5	—	> 30	Moderate

Variance decomposition reveals that 67% of explained variance is shared between $\eta_{\max}$ and K_{eff}, with $\eta_{\max}$ contributing 18% unique and K_{eff} only 3%. Residualized K_{eff} (orthogonalized against $\eta_{\max}$) is significant in only 2/12 conditions. Beta regression does not improve K_{eff} status (3/12). The fundamental issue is the diversity-capability confound ($r = 0.65$): in the current model ecosystem, high-diversity panels necessarily include weaker models.

4.6 Panel Size Sensitivity

Non-independence amplification scales with panel size. As K increases from 3 to 9, each model's participation grows from 91 to 2,002 panels, and SE inflation increases from 3.2× to 7.2× (Table 8).

Table 8: Panel size × correction: $\eta_{\max}$ and K_{eff} significance.

K	N panels	OLS η sig	Boot η sig	SE inflation	Overlap/judge
3	455	11/12	3/12	3.2×	91
5	3,003	11/12	2/12	4.1×	1,001
7	6,435	10/12	1/12	5.8×	3,003
9	5,005	9/12	1/12	7.2×	2,002

K_{eff} bootstrap significance remains at 0–1/12 across all panel sizes. The practical effect size of K_{eff} also diminishes: IQR-based ΔASR drops from 0.8pp (K=3) to 0.2pp (K=9). Larger panels theoretically reduce the weakest-link effect (one weak vote matters less in 9 than in 3), but this cannot be confirmed under corrected inference because non-independence scales faster than signal.

4.7 Aggregation Rule Gradient

While individual predictor significance is fragile, the aggregation rule gradient is robust: it describes R^2 variation, a descriptive statistic unaffected by SE correction (Table 9).

Table 9: Aggregation gradient: OLS vs. bootstrap correction across 4 aggregation rules.

Aggregation	OLS η	OLS K	Boot η	Boot K	R^2
Threshold	12/12	11/12	5/12	2/12	0.89
Majority	11/12	8/12 (7+/1–)	3/12	1/12	0.30
Weighted	9/12	9/12 (9+/0–)	2/12	1/12	0.28
Bayesian	9/12	7/12 (1+/6–)	2/12	0/12	0.10

The R^2 gradient from 0.89 (threshold/unanimity) to 0.10 (Bayesian) is substantial and meaningful: aggregation rule choice determines how predictable panel vulnerability is from composition. Threshold voting maximizes the influence of individual judge weakness; Bayesian calibration minimizes it.

Under OLS, Bayesian aggregation reverses K_{eff} direction (6/7 significant coefficients negative, suggesting diversity becomes protective). However, this reversal does not survive bootstrap correction (0/12 significant), following the same pattern as all other K_{eff} findings. The direction reversal is a directional observation only.

4.8 Panel Selection Strategies

Despite the fragility of regression-based predictor decomposition, the practical recommendation of selecting minimum-$\eta_{\max}$ panels is robust because it depends on panel *ranking*, not on statistical significance of regression coefficients (Table 10).

Table 10: Panel selection strategy comparison (12 conditions, K=3).

Strategy	Win count /12	Mean ASR	vs. Random Δ
Min $\eta_{\max}$	9/12	0.087	–0.085
Two-stage (ours)	9/12	0.084	–0.088
Strongest-K	8/12	0.089	–0.083
Max K_{eff}	2/12	0.142	–0.030
Max family diversity	1/12	0.155	–0.017
Random	0/12	0.172	—

Min-$\eta_{\max}$ and our two-stage algorithm (screen $\eta \leq$ threshold, then maximize K_{eff}) tie at 9/12, with negligible ASR difference (0.003pp). The two-stage algorithm provides no marginal improvement over simply selecting the most individually robust models. This is because individual robustness (η) and model capability are highly correlated ($r = 0.72$): selecting the most robust models $\approx$ selecting the strongest models.

4.9 ASR Concentration Analysis

Per-judge ASR distribution analysis shows that adaptive adversaries produce moderately higher concentration than random faults (Gini 0.28 vs. 0.19, $p = 0.04$, Cohen's $d = 0.22$). However, Gini is primarily driven by capability gap ($r = 0.55$ with $\eta_{\max}$) rather than adversary strategy, and the effect size is small. Targeted attacks show higher concentration than shared-bias attacks (Gini 0.35 vs. 0.22), consistent with the directional taxonomy but insufficient to independently establish an adversary targeting claim.

Table 11: ASR concentration metrics by attack category.

Attack category	Median Gini	Weakest-judge share	vs. RF p
Targeted (PI, SM, PB)	0.35	0.48	0.02
Shared-bias (syc, verb, auth)	0.22	0.38	0.28
All adaptive	0.28	0.42	0.04
Random-fault baseline	0.19	0.36	—

4.10 Open-Ended Generation Benchmark

MT-Bench results mirror the MCQ pattern but with lower R^2 (median 0.15 vs. 0.25 for MMLU). Under bootstrap, $\eta_{\max}$ retains significance in only 1/12 conditions and K_{eff} in 0/12. Cross-benchmark consistency is moderate for $\eta_{\max}$ profiles ($r = 0.52$ MMLU–MT) and weak for K_{eff} ($r = 0.38$).

Table 12: MT-Bench: OLS vs. bootstrap (6 attack conditions). * indicates adapted attack implementation.

Attack	OLS η p	Boot η p	OLS K p	Boot K p	R^2
prompt_inj	0.008	0.22	0.05	0.48	0.18
score_manip	0.015	0.31	0.08	0.55	0.14
sycophancy*	0.042	0.51	0.12	0.63	0.11
verbosity*	0.038	0.48	0.18	0.58	0.09
position_bias	0.003	0.08	0.04	0.38	0.22
authority*	0.055	0.55	0.22	0.68	0.08

The lower R^2 on MT-Bench likely reflects the continuous pairwise comparison format (vs. MCQ's discrete categorical judgments), which introduces more intrinsic scoring noise. Three attacks required substantial adaptation for open-ended evaluation, limiting direct comparability.

4.11 Partial-Knowledge Adversary

We simulate three adversary knowledge levels: full (oracle), partial (family labels only), and zero (random targeting). An ASR gradient exists ($0.142 \rightarrow 0.128 \rightarrow 0.098$, $p = 0.04$), confirming that adversary knowledge matters. Family knowledge provides moderate value ($\Delta\text{ASR} = 0.030$, $p = 0.01$) over zero knowledge, but exact vulnerability profiles add little beyond family information ($\Delta = 0.014$, $p = 0.18$). Neither $\eta_{\max}$ nor K_{eff} significance improves at any knowledge level.

Table 13: Adversary knowledge level vs. panel vulnerability.

Knowledge	Mean ASR	Boot η sig	Boot K sig	R^2
Full (oracle)	0.142	3/12	0/12	0.25
Partial (family)	0.128	2/12	0/12	0.18
Zero (random)	0.098	1/12	0/12	0.08

5 Discussion

The Non-Independence Problem Is Structural. Our results demonstrate that combinatorial panel designs create a fundamental tension: exhaustive enumeration maximizes statistical coverage but introduces severe non-independence. Each of N models appears in $\binom{N-1}{K-1}$ panels, creating cross-observation correlation that inflates standard errors by 3.2–7.2× depending on K. This is not a correctable nuisance—it is an intrinsic property of the experimental design.

Neither Predictor Is Reliably Established. Under corrected inference, $\eta_{\max}$ retains significance in only 3/12 conditions (bootstrap) and K_{eff} in 1/12. The VIF analysis (> 7 for both) confirms that 67% of explained variance is shared, with K_{eff} contributing only 3% unique variance. The regression-based decomposition framework is fundamentally inadequate for isolating independent predictor effects with 15 model clusters.

What Survives. Two findings are robust. First, the aggregation R^2 gradient (0.89 to 0.10) is descriptive and independent of inference correction—it characterizes how much panel composition matters under different voting rules. Second, minimum-$\eta_{\max}$ selection dominates alternative strategies in 9/12 conditions, a result based on direct comparison rather than regression significance.

The Diversity-Capability Confound. The correlation between $\eta_{\max}$ and K_{eff} ($r = 0.65$, $R^2 = 0.42$) is primarily driven by cross-tier differences ($r = 0.72$ for cross-tier panels vs. $r = 0.22$–0.35 within-tier). In the current LLM ecosystem, capability predicts robustness: high-diversity panels necessarily include weaker models. This confound is not an artifact of our analysis but a structural feature of the model landscape.

Implications for Evaluation Practice. Our results suggest three practical takeaways: (1) prioritize individual judge quality over panel diversity; (2) choose aggregation rules deliberately—Bayesian calibration reduces composition sensitivity while threshold voting amplifies it; (3) be cautious about regression-based conclusions from combinatorial panel designs without non-independence correction.

6 Conclusion

We conducted a large-scale study of LLM evaluation panel vulnerability, finding that standard OLS regression dramatically overstates predictor significance in combinatorial panel designs due to structural non-independence. Under wild cluster bootstrap correction, $\eta_{\max}$ significance drops from 11/12 to 3/12 and K_{eff} from 8/12 to 1/12, with SE inflation of 3.2×. This collapse is itself a methodological contribution: it provides a concrete demonstration of why non-independence correction is essential for combinatorial evaluation research.

Two findings survive: aggregation rule choice modulates vulnerability predictability (R^2 from 0.89 to 0.10), and minimum-$\eta_{\max}$ panel selection outperforms alternatives in 9/12 conditions. We release a 15-model, 455-panel adversarial evaluation dataset as a resource for future work.

Limitations

Our study has three main limitations. First, with 15 models and therefore 15 clusters, bootstrap correction is conservative and may be overly so—the true significance of $\eta_{\max}$ likely lies between the OLS upper bound (11/12) and the bootstrap lower bound (3/12). Studies with larger model pools ($N \geq 30$) would provide more reliable corrected inference. Second, our findings are primarily established on MCQ benchmarks (MMLU, ARC); MT-Bench results show limited transfer, suggesting that the vulnerability dynamics may be format-dependent. Third, the diversity-capability confound ($r = 0.65$) is a property of the current model ecosystem and may change as the landscape evolves.